HYBRID PARTICIPATION INNOVATION LAB · ARTICLE 3

The valuable plurality of the citizen sciences

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THE PAPER AT A GLANCE

Is "just collecting data" really second-class participation?

THE COMMON VIEW

Most citizen science is **contributory**: citizens gather data for experts. Critics say this is not inclusive, transformative or democratic. Co-creation is presented as the ideal.

THE AUTHORS' ANSWER

Contributory projects can also be **emancipatory**: they can change policy, change behaviour and reach hard-to-reach groups. The future is a **plurality** of citizen sciences.

THREE MEANINGS OF CITIZEN SCIENCE

From data gatherers to co-researchers

Contributory

Volunteers collect data; experts design and interpret. Tradition of amateur science (Bonney).

Participatory

Science opened up to wider publics and done by citizens themselves (Irwin, 1995).

Co-creative

Citizens take part in every stage, from the research question to the analysis (Shirk et al., 2012).

THE CRITIQUES

Unpaid labour serving science or industry; experiential knowledge ignored; and in the US, participants are mostly white, above-median income, college-educated.

THE PROBLEM WITH RANKING

The participation ladder makes a hierarchy



Arnstein's ladder (1969), in simplified form

Why the authors push back

Treating co-creation as the top rung makes contributory projects look "less than", and this shapes expert judgement and funding.

But not everyone has the time, resources or commitment to co-create. And any participatory engagement raises questions of strategy and power.

THREE CONTRIBUTORY CASES

Simple data collection, real impact

FLANDERS · 2018

CuriousNoses

20,000 people measured NO2 air pollution in their street for a month.

Impact: air quality became an election issue; a Brussels court accepted the data as indicative evidence; people changed how they commute.

UGANDA · 2019

ATRAP

25 trained volunteers reported snails that spread parasitic diseases, weekly, by phone app.

Impact: warning signs at risky sites; volunteers called "Snail Doctor"; one entered local politics; copied in Chad.

NEW ZEALAND · 2016–

Flip the Fleet

645 electric-car drivers share monthly data on distance, charging and efficiency.

Impact: better-informed buyers and a public debate on EV policy; it is slowly moving toward co-creation.

THE CORE ARGUMENT

Three ways to engage, all needed

Deliberation

Listening, questioning your own assumptions, learning together.

Participation

Mobilising like-minded people to have a voice in decisions.

Representation

Letting spokespersons and data speak on citizens' behalf in the political arena.

Low-threshold tools (cheap, simple, autonomous) let minorities and busy people join at scale, and scale makes institutions listen.

A PLEA FOR PRAGMATISM

Start from the problem, not the format

ASK FIRST

1. What is the problem or challenge?
2. For whom, and why?
3. What change: scientific, societal or systemic?

RISKS TO MANAGE

Powerful actors using volunteers' data "under the guise of opening and democratizing science".

Contribution turning into a one-way, "tick-box" consultation.

Citizens doing routinised labour for economic reasons.

OUR CRITICAL READING

A useful correction, built on success stories

Strengths

Challenges a normative bias that shapes funding and evaluation.

Concrete impacts: a court case, elections, new local leaders.

Practical questions to choose the right format.

Limits

Authors' own experience as evidence, and only successful ones.

No data on who took part: did low thresholds really include minorities?

Impact depended on media and allies, not only on the format.

WHAT WE TAKE FOR OUR PROJECT

A two-track citizen science design

TRACK 1 · WIDE

Every household contributes

A simple energy postcard or a smart plug: low effort, reaches busy and non-digital tenants, gives the scale that makes ATER and the City listen.

TRACK 2 · DEEP

Volunteers co-create

A smaller group designs the protocol, interprets the data with researchers and proposes changes to the AI and the fairness rule.

Plus: validated methods for credibility, and recognised "energy ambassadors", like the Snail Doctors.

FOR DISCUSSION

Questions for the class

1. Is reaching 20,000 people as data collectors more democratic than 20 people co-creating?

2. When does contributing data become unpaid labour?

3. How does this paper challenge the professor's rule that citizens must take part in interpretation?